Forecasting Bakery Sales with Weather and Economic Parameters

YAP470 - Fall 2024 Interim Report

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***Abstract*—In order to assist bakeries in forecasting their daily and monthly sales, this study suggests a predictive system. Bakeries find it difficult to forecast their sales due to the strong demand for fresh goods. The proposed method integrates weather and economic variables using various learning techniques to generate precise sales forecasts. Additionally, the scope of this study includes weather forecast and oil price forecast to be used in Bakery sales prediction. For weather forecasting, LSTM model was trained using dataset prepared with data obtained from Metostat library, GRU model was trained using dataset prepared with data obtained from yfinance library for oil price prediction, Multiple Linear Regression model was trained using dataset prepared with preprocess of datasets obtained from Kaggle for bakery sales prediction. In this part of the study, the preparation process of the datasets, some analyzes related to the datasets, the outputs obtained from the models and the RMSE, MAPE, R^2 and Adjusted R^2 scores that evaluate the performance of the models are included.**

The source code and datasets for this study are available at: https://github.com/tunakodal/Forecasting-Bakery-Sales.git

***Index Terms*—bakery sales forecasting, weather forecasting, economic forecasting, LSTM, GRU, Multiple Linear Regression**

I. Introduction

FAO(Food Wastege Footprint) estimates that the world loses or wastes nearly one-third of the food produced for human consumption. The fact that food loss and waste are happening at a time when food prices are rising and many people's food insecurity is growing only increases the worry. The environment is negatively impacted by food losses and waste because they contribute to greenhouse gas emissions and waste energy and production factors [1].

According to estimates, restaurants, convenience stores, and supermarkets alone discard at least 35 million tonnes, or 77 billion pounds, of edible food annually in the US, valued at almost $30 billion. According to a related study, 26% of the food that convenience stores stock ends up in the trash. According to another study, 14% of the food that American households buy ends up in the trash, costing a family of four almost $600 annually and contributing $43 billion to the American economy. Additionally, 12–14% of all municipal solid waste is dumped in landfills. In total, the issue of food waste costs the US economy about $75 billion a year. In addition to this enormous loss, Americans spend $1 billion disposing of their food waste, according to the Environmental Protection Agency [2].

Focusing on bakery and other confectionery products is necessary because, aside from fruit and vegetables, this product category is the most frequently wasted. Retailers prioritize their availability on shelves and a high degree of diversity over other sales aspects in order to increase customer satisfaction. However, this policy causes more food waste, which in turn causes more environmental harm. As a result, there are financial losses and social pressures because edible food is thrown away [3].

The food supply chain is complex and interconnected, with numerous producers, processors, retailers, and consumers all interacting with other actors in the chain. This leads to a more complex and connected economics of food loss and waste [4].

The goal of this project is to provide an alternative approach to situations where bakeries are unable to effectively organize and forecast their sales by utilizing machine learning and deep learning techniques. It is anticipated that the method mentioned will be able to forecast sales based on historical sales trends, current weather patterns, and the state of the economy. The improvement of bakeries' scalability production with short-term forecasts and supply forecasts with long-term forecasts are the main objectives of this study. In order provide these forecasts features, weather and economic conditions will be predicted before short- and long-term sales predictions. Because it is believed that making weather forecasts in the long term will yield poor results, it has been used only for short-term predictions; sales trends and economic forecasts

*Both authors contributed equally to this work.

have been used for both short- and long-term predictions.

The datasets and techniques used to make these estimates will be thoroughly described in the Methodology section. The study's objective is to produce precise estimates based on the metrics that will be thoroughly explained in the Methodology section for bakery sales in short and long term, using the weather and oil price forecasting, as well as using the past demand datasets.

A. Overview

- The goal of the project and the arrangement of the sections in this document are made clear by the project's introduction, the first phase. This section will serve as the project's introduction.

- A comprehensive review of the literature and studies that are comparable to this study are included in the second section. These studies have made significant advances in weather, economic, and bakery sales forecasting. In this section, studies from the literature are introduced and discussed, with particular attention to how they differ and how they are similar to the study will be conducted.

- The study's methodology, datasets, and evaluation metrics will be covered in the third section. The datasets used for each of the three approaches will be covered initially in the first part of the third section. The datasets will be described, and the contents of the data will be mentioned. The models will be introduced, along with their architectural structure, areas of use, and general details about the areas in which they are successful, in the second part of the section. The evaluation metrics that will be used to assess the outputs from all three models will be presented in the final part, along with a discussion of the error calculation techniques and evaluation techniques used by these studies.

- Three headings will comprise the fourth section's Features title, which will offer a thorough analysis of the features employed in each technique. The features in the dataset used for weather forecasting will be described, along with their impact, in the first part of the fourth section. The features of the economic forecasting dataset will be described, along with their impact, in the second part of the fourth section. The features of the bakery sales forecast datasets will be described, along with their impact, in the third part of the fourth section.

- Fifth, the section on Experimental Results concentrates on the implementation of the approach this project presents and provides a step-by-step explanation of how to accomplish. Information about which portion of the datasets used is organized and how is provided in the first part of the fifth section. Additionally, this section explains how these datasets are ultimately combined. Comprehensive manual analyses and some probability and statistical analyses of the prepared datasets and their features are presented in the second part of the fifth section. The outputs from the models under specific conditions, in addition to the error values of these outputs, are provided in the third part of the fifth section.

- Results from the study are discussed and evaluated in the sixth section, which is a discussion section.

- The study's conclusion and a summary of the findings are included in the seventh section.

- The future studies section, which is included in the eighth section, contains recommendations for future research projects as well as projects that aim to expand on, increase, or utilize this study. The techniques that will be used to increase model accuracy and that are not within the scope of this investigation but are believed to increase model accuracy will be covered in this section. Additionally, the studies planned to be carried out as a continuation of this study are also included in this section.

- The references section contains all of the sources that were used in the research.

II. Related Work

A. Weather Forecasting

1) Jain and Mallick, gathered the data data from a weather station was used to train an ARIMA model. A model commonly employed in time series forecasting was applied in a study similar to this technique. Alternative approaches that focused on the ARIMA model's capabilities were proposed, in contrast to this study, no evaluation was included because testing on this dataset would be difficult. Furthermore, the preparation steps taken on the data and the format of the data were not included in the study [5].

2) Babu et. al., are developing an LSTM model using data containing various features from 5 cities via Weather Underground. Approximately 6.5 years of data was used in their study. Since they collected the train and test data according to time, they divided them according to time. Unlike this study, data preprocessing operations and data analysis were not included. They proposed a new method called T-LSTM, in which each training data point's impact on the parameters is determined by how similar it is to the test data point. In addition, just the Adam optimizer was used to run the train [6].

3) By Salmana et. al., single-layer and multi-layer LSTM models were employed to forecast the weather at Indonesia's Hang Nadim Airport, and the impact of variables including temperature, pressure, and humidity on forecast accuracy was investigated. Despite some similarities to the single-layer LSTM model employed in this study, the multi-layer LSTM model was applied in a different way. When the pressure variable was used, the multi-layer LSTM model produced the best results. Preprocessing techniques like normalization, rescaling, and smoothing using moving average were used during the data preparation phase [7].

B. Oil Price Forecasting

1) Ahmed and Shabri, West Texas Intermediate (WTI) crude oil spot price data collected for approximately 30 years were used to train ARIMA, GARCH and SVM models using normalization. Although the training was performed on different models from this study, the evaluation of the models with RMSE(Root Mean Squared Error) similarly and MAE(Mean Absolute Error) differently error measures was applied. In addition, WTI(West Texas Intermediate) oil data was used similarly, but Brent oil data was not used differently [8].

2) Wu et. al., concentrated on using WTI (West Texas Intermediate) data to train the EEMD-LSTM model, which combines the LSTM (Long Short-Term Memory) and EEMD (Ensemble Empirical Mode Decomposition) techniques. Similar to this study, the evaluation criteria which is used were Mean Absolute Percentage Error (MAPE) and Root Mean Squared Error (RMSE) [9].

3) Hamayel et. al., They trained and evaluated LSTM, bi-LSTM, and GRU (Gated Recurrent Unit) models. Similar to this study, a study was conducted using the GRU model to predict the future prices of BTC, ETH, and LTC cryptocurrency data collected via MarketWatch. GRU was found to have the highest accuracy value based on the Mean Absolute Percentage Error (MAPE) and Root Mean Squared Error (RMSE) metrics, which were used in the study similar to this [10].

C. Bakery Sales Forecasting

1) Acı and Yergök, used consumption data obtained from the cafeteria of Mersin University and trained different methods such as Artificial Neural Network (ANN), Gaussian Process Regression (GPR), Support Vector Regression (SVR), Regression Tree, and Ensemble Decision Tree (EDT) with the intention of making predictions to reduce food waste in order to serve the same purpose as this study. Different models were included under these methods in the study. Similar to this study, a linear regression model was also included in the study. Evaluation metrics such as Mean Absolute Error (MAE), Mean Square Error (MSE) and Multiple Correlation Coefficient (R) were used. As a result, it was concluded that Ensemble Decision Tree (Boosted EDT) performed better than others [11].

2) Lindström, using the time series data from Coop Värmland, trained models for Decision Tree Regression, Random Forest Regression, Artificial Neural Networks(ANN), Recurrent Neural Networks(RNN), and Prophet (a time series model). The data contains elements like sales, product category, and time information, and the study includes the analysis of these data. In contrast to this study, their study focuses on forecasting sales of bread and coffee. The study's overall findings are presented in detailed using a number of evaluation metrics, including MAE, MAPE, R2/R2adj, and MSE similarly [12].

3) Suryawan1 et. al., used sales data from Bandung White Bread Factory to train ARIMA, LSTM, and Prophet models. After evaluating the models, the MAPE, MSE, and RMSE is used as evaluation metrics, it was found that ARIMA(1,0,2) performed better than the other models. (ARIMA(1,0,2); is the scenario in which AR = 1, I = 0, and MA = 2 are the model parameters.) The ARIMA method (1,0,2) generated the lowest error accuracy value, with an RMSE of 47.4139, an MSE of 2248.0822, and a MAPE of 4.548% [13].

III. Methodology

An in-depth look at the datasets, models and evaluation metrics planned to be used in this study will be presented. The pipeline of the study is demonstrated in Figure 1.

During all random splits in this study, random states were used as 42.

A. Datasets

The StandardScaler operation was applied to standardize the data. This operation Standardize features by removing the mean and scaling to unit variance. [14] Certain models, particularly gradient-based models, may suffer when the ranges of various features in datasets differ. Because of this, standardization is applied.

1) Weather Forecasting Dataset: Weather Forecasting Dataset: The Meteostat Python library offers a straightforward API for accessing public weather and climate data. Meteostat collects historical observations and statistics from various public interfaces. The majority of the interfaces are governmental. Data sources include national weather services such as the

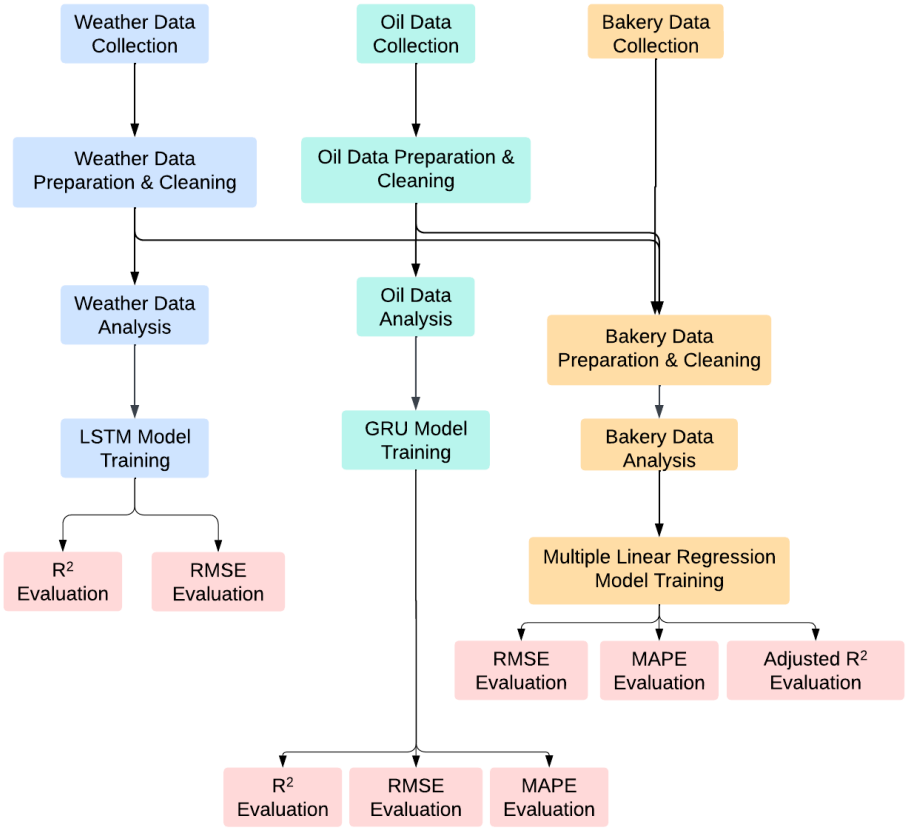


Fig. 1. Flowchart of the study completed up to this stage

National Oceanic and Atmospheric Administration (NOAA) and Germany's national meteorological service (DWD) [15].

Data for each date was obtained using the date and location information. Edinburgh was used as the location to be compatible with the bakery dataset. The none(nan) prcp, snow, wdir, wpgt, tsun columns in this data were not used:

- prcp: precipitation, the amount of rainfall in millimetres.
- snow: the amount of snow in millimeters.
- wdir: wind direction, is determined in degrees.
- wpgt: wind gust, or maximum wind speed in km/h.
- tsun: sunshine, the amount of hours of sunshinet.

In addition, the dates for which some data were missing or corrupted were removed from the dataset. As a result, a dataset containing 3213 rows of tmin, tmax, tavg, wspd, pres features between the dates 2015.09.07 - 2024.11.08 was obtained in Edinburgh. Features will be explained in features section in detailed. This dataset was randomly divided into three groups: 80% train, 10% validation and 10% test. Following the random division, it was determined that there was no clustering of close dates in the train, validation or test datasets.

An example from the dataset: [1]

time	tmin	tmax	tavg	wspd	pres
2016-03-14,	0.0	10.0	5.3	9.1	1037.0

a) Exploratory Data Analysis:

Weather Data Over Time The weather data over time (2015-2024) is illustrated in Figure 2.

[1]The study used the date 14.03.2016 as an example to refer to the day that Google's AlphaGO artificial intelligence model defeated Lee Sedol, marking a significant milestone in the artificial intelligence history.

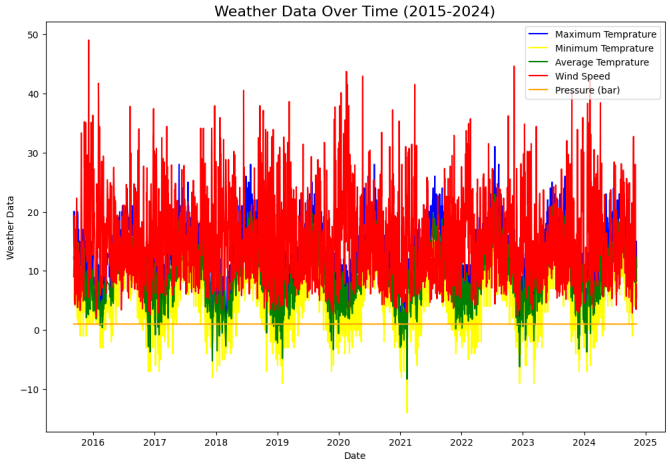


Fig. 2. Weather Data Over Time

The scikit-learn preprocessing package's StandardScaler was used to standardize the dataset:

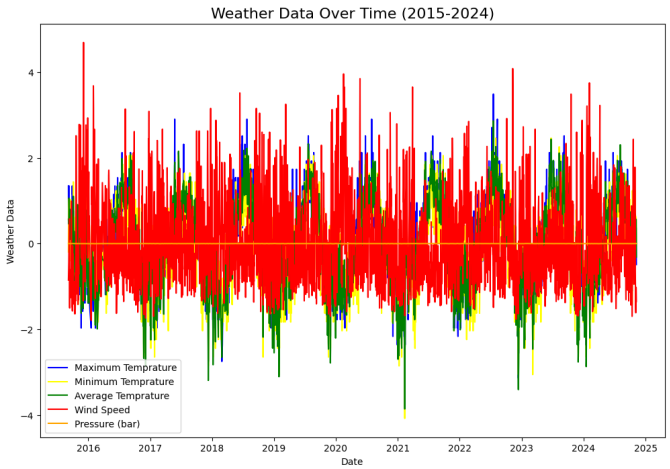


Fig. 3. Normalized Weather Data Over Time

2) Oil Price Forecasting Dataset: Yahoo Finance is a media goods within the Yahoo network. It offers financial news, data, and commentary, such as stock quotes, press releases, financial reports, and original content [16]. yfinance provides a threaded Python-based way to download market data from Yahoo!® finance [17]. Brent oil and West Texas Intermediate oil data were used from yfinance library. Crude oil extracted in the North Sea between Norway and England (United Kingdom) is called Brent oil. Brent Oil has an important place in determining prices in the world oil market and is used as a reference point [18]. West Texas Intermediate (WTI) is the main oil benchmark for North America because it is sourced in the United States. Brent crude and WTI crude are the most common benchmarks, but their prices are frequently compared. The difference in price between Brent and WTI is known as the Brent-WTI spread [19].

To obtain a more stable and generalizable approach, the average of these two data points was calculated and added to the data as a column, which was then used. The days that could not be obtained because the stock market was closed during the data collection were excluded from the dataset in this section. This dataset was randomly divided into three groups: 80% train, 10% validation and 10% test.

A dataset is obtained with 4225 rows of date, Brent price, WTI price, average price features between the dates 2000.08.23 - 2024.07.29 for WTI oil, 2007.08.30 - 2024.07.29 for Brent oil, 2007.08.30 - 2024.07.29 for average oil price.

An example from the dataset:

Date	Brent Price	WTI Price	Average Price
16-03-14	39.52999877929688	37.18000030517578	38.35499954223633

a) Exploratory Data Analysis:
Oil Prices Over Time The average oil price over time (2000-2024) is shown in Figure 4.



Fig. 4. Oil Prices Over Time

The scikit-learn preprocessing package's StandardScaler was used to standardize the dataset:



Fig. 5. Normalized Oil Prices Over Time

3) Bakery Sales Forecasting Datasets: In this study, the Bakery Sales Dataset(Edinburgh) [20] dataset and Bakery Sales(Korea) [21] dataset on Kaggle [22] were used.

An example from Bakery Sales Dataset(Edinburgh) dataset:

TransactionNo	Items	DateTime	Daypart	DayType
691	Bread	2016-06-11 12:22:28	Afternoon	Weekend

An example from Bakery Sales(Korea) dataset:

datetime	day_of_week	place	americano	croissant	...	orange pound	cheese cake	lemon ade	total
2019-10-31 12:18	Thur	후평동		1		1			14800

Although these two datasets were prepared to serve the same purpose, they were very different in terms of format. In order to eliminate these differences, the Bakery Sales(Korea) dataset was converted to Bakery Sales Dataset(Edinburgh) dataset in terms of format. A dataset was obtained by combining the edited Bakery Sales(Korea) dataset with the Bakery Sales Dataset(Edinburgh) dataset. The items sold in these datasets were listed and the items that were sold were examined. Item sales such as tshirt, valentine's card, postcard, nomad bagthat etc. were thought to have no relation to the bakery datasets were removed from the dataset. In addition, like duck egg(12), honey(5), bacon(1), tiramisu (1) etc. products that were observed to have sold less than 40 items throughout the entire dataset and that could not be strongly associated with the bakery were removed from the bakery dataset. (The number of times the removed products appear in the dataset is given in parentheses.)

Then, products that were thought to be the same product despite being expressed differently due to some regional name differences and products with very minor differences between them were combined to form a single product class. The combinations made:

- Crossiant, medialuna, pain au chocolat, almonde crossiant, tiramisu crossiant products were collected under the title of crossiant.
- Latte, vanilla latte, cafe latte products were collected under the title of latte.
- In addition, while there were sub-types of coffee such as americano and latte in the dataset, there were also items specified only as coffee. In order not to create the misconception that this coffee would cover other coffees, the coffees mentioned only as coffee were expressed as filter coffee. In this way, the bakery related part of the dataset preparation was completed.

New features thought to be related to bakery were added to this dataset. The first of these added features was the isHoliday feature, which shows the holidays. A day being a holiday was determined according to the:

Edinburgh:

- 22nd October: Republic Day
- 27th October: Republic Day
- 20th December: World Solidarity Day
- 25th December: Christmas
- 26th December: Boxing Day
- 1st January: New Year
- 2nd January: New Year Holiday
- 19th January: National Education Day
- 25th January: Burn's Night (Literary Day)

Korea:

- 1st January: New Year's Day
- 7th February: Seollal (Korean New Year)
- 8th February: Seollal (Korean New Year)
- 9th February: Seollal (Korean New Year)
- 10th February: Seollal (Korean New Year)
- 1st March: Independence Day
- 1st May: Labor Day
- 5th May: Children's Day
- 14th May: Mother's Day
- 6th June: Memorial Day
- 14th September: Chuseok (Korean Harvest Festival)
- 15th September: Chuseok (Korean Harvest Festival)
- 16th September: Chuseok (Korean Harvest Festival)
- 9th October: Hangul Day
- 25th December: Christmas

The isHoliday feature was determined as 1 if it was a holiday in that place and 0 if it was not a holiday. Weekends are not included in this isHoliday feature because the day feature (Monday, Tuesday ...) will already be given to the model.

TransactionNo	Items	Daypart	Day	Place	Date	Time	isHoliday
1	coffee	Morning	Monday	Edinburgh	2016-10-31	08:28:31	0

Then, in order to prepare the weather data, the latitude and longitude coordinates of the locations in the Korean and

Edinburgh bakery datasets were added to the dataset as a feature.

An example from dataset:

TransactionNo	Items	Daypart	Day	Place	Date	Time	isHoliday	Latitude	Longitude
1	coffee	Morning	Monday	Edinburgh	2016-10-31	08:28:31	0	55.9533456	-3.1883749

Using these coordinates, weather data for the relevant dates were obtained from the metostat library in Python. This data includes tmin, tmax, tavg, wspd, pres features for each date. Due to the data in the library, some of the dates in the bakery dataset have missing weather information. Bakery data, which we plan to use as a feature, was removed from the dataset because it was thought that accurate weather data could not be obtained due to the lack of this data. Detailed description of the library is given in the weather prediction dataset section.

An example from dataset:

Then, in order to add the economic conditions under which the existing sales were made to the dataset, Brent oil and WTI oil data on the date of the sales were collected with the yfinance library. Since no stock market data was created on the days when the stock market was closed, it was aimed to create a complete dataset by filling the gaps created by the stock market closure with the stock market data that existed just before. The details of the Yahoo finance library are given in the oil price dataset section. Brent price, WTI price features were taken from this library and added to the dataset. Since this data was used in the study to provide an overview of the general economic situation, the average of Brent oil and WTI values was used, too.

An example from dataset:

After the dataset was prepared in this way, since the project aimed to produce on a daily and monthly basis, in order to obtain a format focused on daily sales, the sales made on the same day were arranged in a way that the weather, oil prices and isHoliday features were represented in a single line without any quality loss. Since the Bakery Sales Dataset(Edinburgh) and Bakery Sales(Korea) datasets do not have the same dates, there was no change in the location and location-related features of the sales collected under the same day.

An example from dataset:

As a result, this dataset prepared has become a dataset consisting of 1554 rows and day, place, date, isHoliday, Brent price, WTI price, avg_oil_price, latitude, longitude, tmin, tmax, tavg, wspd, pres features between the dates 2016.11.31 - 2020.04.29. There are dates between these dates where there are no transactions. Since the dates of the two datasets were obtained at different times, there is no sales data for any date around 2018. It has been randomly divided as 80% train, 10%

TransactionNo	Items	Daypart	Day	Place	Date	Time	isHoliday	Latitude	Longitude
1	coffee	Morning	Monday	Edinburgh	2016-10-31	08:28:31	0	55.9533456	-3.1883749
min	**tmax**	**tavg**	**wspd**	**pres**					
8.0	14.0	10.7	11.2	1026.0					

TransactionNo	Items	Daypart	Day	Place	Date	Time	isHoliday	Latitude
1	coffee	Morning	Monday	Edinburgh	2016-10-31	08:28:31	0	55.9533456
Longitude	**min**	**tmax**	**tavg**	**wspd**	**pres**	**Brent_Price**	**WTI_Price**	**Avg_Oil_Price**
-3.1883749	8.0	14.0	10.7	11.2	1026.0	48.29999923706055	46.86000061035156	47.579999923706055

validation and 10% test data. Following the random division, it was determined that there was no clustering of close dates in the train, validation and test datasets.

a) Exploratory Data Analysis:

Oil Prices Over Time (2016-2017) The oil prices over time (2016-2017) is shown in Figure 6.

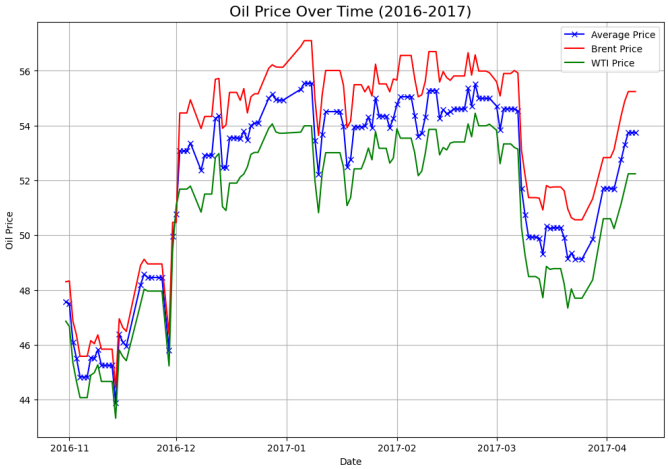


Fig. 6. Oil Prices Over Time (2016-2017)

The scikit-learn preprocessing package's StandardScaler was used to standardize the dataset:

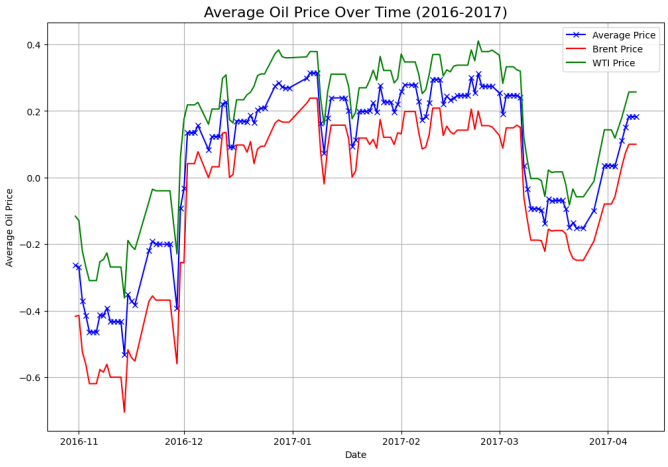


Fig. 7. Normalized Oil Prices Over Time (2016-2017)

Oil Price Over Time (2019-2020) The oil prices over time (2019-2020) is presented in Figure 8.

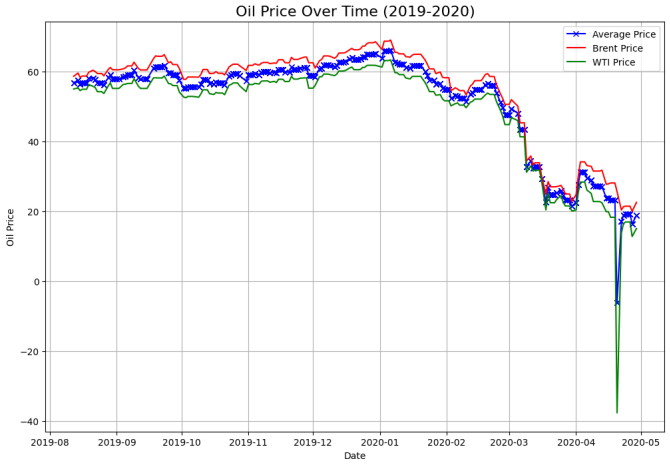


Fig. 8. Oil Price Over Time (2019-2020)

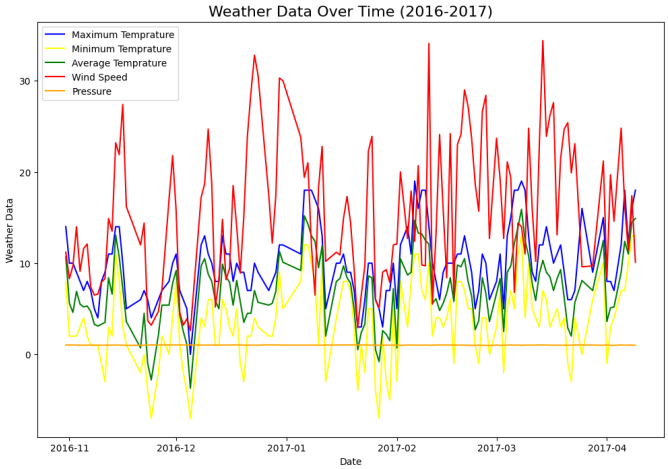


Fig. 10. Weather Data Over Time (2016-2017)

The scikit-learn preprocessing package's StandardScaler was used to standardize the dataset:

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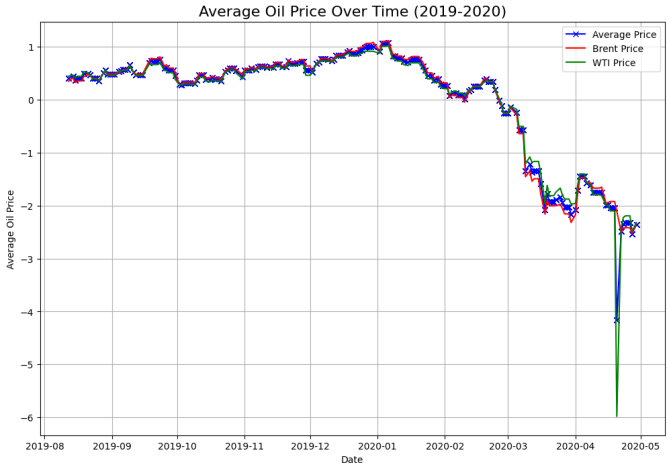


Fig. 9. Normalized Oil Prices Over Time (2019-2020)

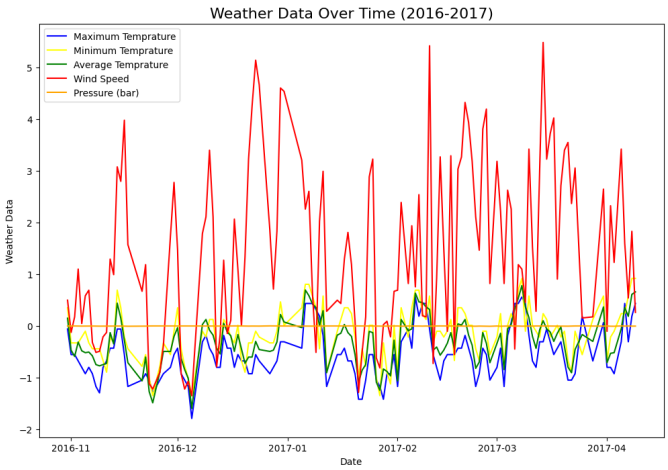


Fig. 11. Normalized Weather Data Over Time (2016-2017)

Weather Data Over Time (2016-2017) The weather data over time (2016-2017) is illustrated in Figure 10.

Weather Data Over Time (2019-2020) The weather data over time (2019-2020) is depicted in Figure 12.

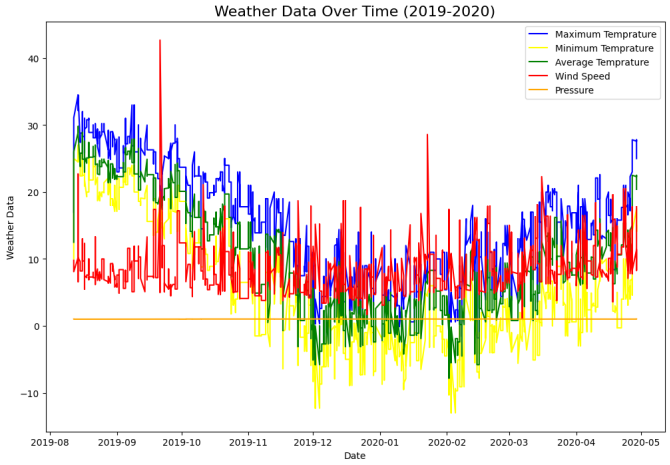


Fig. 12. Weather Data Over Time (2019-2020)

The scikit-learn preprocessing package's StandardScaler was used to standardize the dataset:

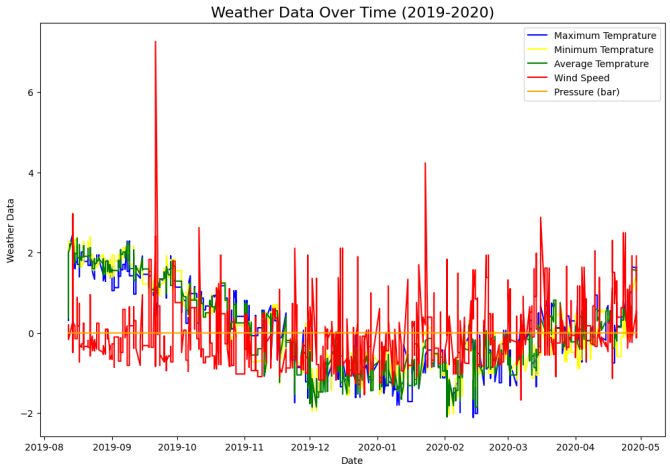


Fig. 13. Normalized Weather Data Over Time (2019-2020)

Cumulative Demands of Items The cumulative total of item features in this dataset is given in Figure 14.

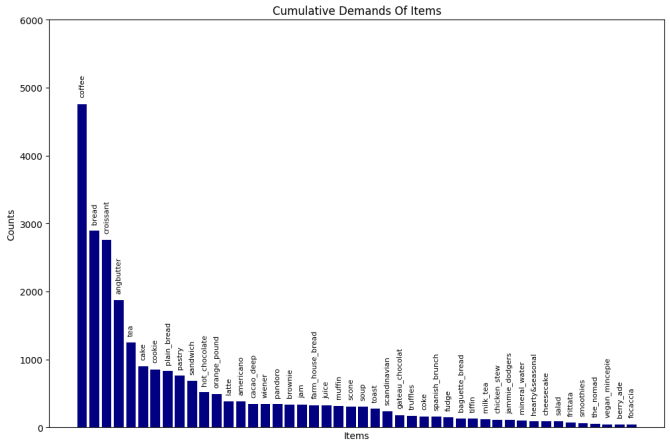


Fig. 14. Cumulative Demands Of Items

The scikit-learn preprocessing package's StandardScaler was used to standardize the dataset:

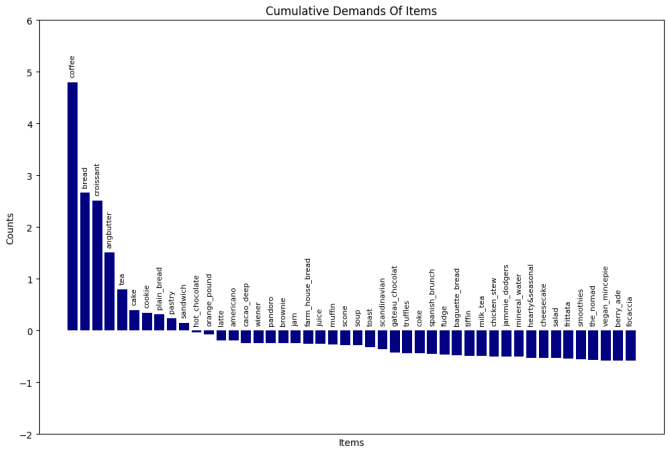


Fig. 15. Normalized Cumulative Demands Of Items

B. Models

1) Weather Forecasting Model: In this study, LSTM(Long Short-Term Memory) model will be used to predict the weather [23].

Recurrent Neural Networks with Long Short-Term Memory (LSTM) are an efficient and scalable model for a variety of sequential data-driven learning problems. Previous approaches to addressing these issues were either problem-specific or failed to successfully scale to long-term dependencies. Long-term temporal dependencies, however, can be effectively captured by LSTMs, which are both general and efficient.

The LSTM architecture is based on a memory cell that can maintain its state over time, as well as nonlinear gating units that control information flow into and out of the cell. LSTMs are now applied to many learning problems, which differ significantly in scale and nature from the problems that these improvements were originally tested on [24].

In this study, the train learning rate was used as default (0.001), batch size was 32, epoch number was 50, activation function was ReLU and loss function was Huber loss.

2) Oil Price Forecasting Model: In this study, GRU(Gated Recurrent Unit) model will be used to predict oil prices. [25]

The GRU is a simplified version of the LSTM. GRU's internal structure is simpler than LSTM, making it easier to train due to fewer computations. These simplifications are achieved primarily through two methods: (1) The input and forget gates are merged into a single gate known as the update gate. (2) The internal/cell and hidden states are merged [26].

In this study, the train learning rate was used as default (0.001), batch size was 32, epoch number was 50, activation

function was ReLU, and loss function was Mean Squared Error (MSE).

3) Bakery Sales Forecasting Models: In this study, Multiple Linear Regression model will be used to predict bakery sales.

Linear regression is a modeling technique that analyzes data and makes predictions. In simple linear regression, a bivariate model is constructed to predict a response variable (y) from an explanatory variable (x). Multiple Linear Regression extends the model to include multiple explanatory variables (x1,x2,. . . .,xp), creating a multivariate model. To fit a straight line to the data points, linear regression is what we use to make predictions [27].

In this study, the train default solver was used as 'auto', and data normalization was applied with StandardScaler.

C. Evaluation Metrics

1) Weather Forecasting Evaluation Metrics:

a) Root Mean Squared Error (RMSE):
It calculates the average difference between values predicted by a model and actual values. It gives an estimate of how well the model can predict the target value (accuracy) [28].

b) R^2 Score:
R-squared (R^2) is a metric that measures how well the independent variables in a statistical model explain the variation in the dependent variables. It ranges from 0 to 1, with 1 indicating a perfect fit between the model and the data [29].

c) Mean Absolute Percentage Error (MAPE):
Mean absolute percentage error is a relative error measure that uses absolute values to keep positive and negative errors from canceling each other out and uses relative errors to allow you to compare forecast accuracy between time-series models [30].

2) Oil Price Forecasting Evaluation Metrics:

a) Root Mean Squared Error (RMSE):
Its definition is given in the Weather Forecasting and Oil Price Forecasting Evaluation Metrics section.

b) R^2 Score:
Its definition is given in the Weather Forecasting and Oil Price Forecasting Evaluation Metrics section.

3) Bakery Sales Forecasting Evaluation Metrics:

a) Root Mean Squared Error (RMSE):
Its definition is given in the Weather Forecasting and Oil Price Forecasting Evaluation Metrics section.

b) Mean Absolute Percentage Error (MAPE):
Its definition is given in the Weather Forecasting and Oil Price Forecasting Evaluation Metrics section.

c) Adjusted R^2 Score:
The adjusted R^2 is a corrected goodness-of-fit (model accuracy) metric for linear models. It calculates the percentage of variance in the target field that can be explained by one or more inputs. R^2 tends to overestimate the fit of the linear regression. It always increases as the number of effects in the model grows. Adjusted R^2 attempts to compensate for this overestimation. Adjusted R^2 may decrease if a particular effect does not improve the model [31].

IV. Features

In this section, correlation matrices were included for the datasets in order to visualize the linear relationships between the features in the dataset. Thanks to these matrices, the relationships between the features will be easier to understand and data analysis was performed easily.

A. Weather Forecasting Features

1) time: the date to which the weather data in the row belongs

2) tmin: is the minimum temperature value for the date in that row.

3) tmax: is the maximum temperature value for the date in that row.

4) tavg: is the average temperature value for the date in that row.

5) wspd: is the average wind speed value for the date in that row.

6) pres: is the average pressure value for the date in that row.

a) Correlation Matrice:
In all correlation matrices used in this study, red tones represent positive correlation and blue tones represent negative correlation which is shown in the right parts of the figures.

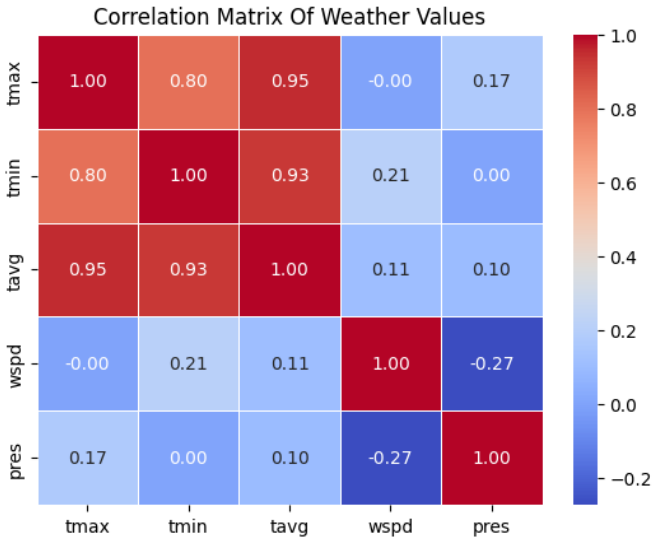


Fig. 16. Weather Forecasting Features Correlation Matrice

As can be seen from this matrix, tmin, tmax, tavg are actually quite positively correlated with each other by nature.

All 3 features indicate air temperature. However, it can be seen that tmin and tmax show the lowest positive correlation between temperatures. This is expected, three features with high correlations were used together in the study to create more precise data. While it can be said that wind speed is negatively correlated with pressure, wind speed and pressure are positively correlated with temperature, but this correlation is not very strong.

B. Oil Price Forecasting Features

1) Date: which day the oil prices obtained in the row belong to

2) Brent Price: the Brent oil price for the date in that line.

3) WTI Price: the West Texas oil price for the date in that line.

4) Average Price: the arithmetic average of the Brent oil price and West Texas Intermediate oil price for the same date which is presented on the row.

a) Correlation Matrice:

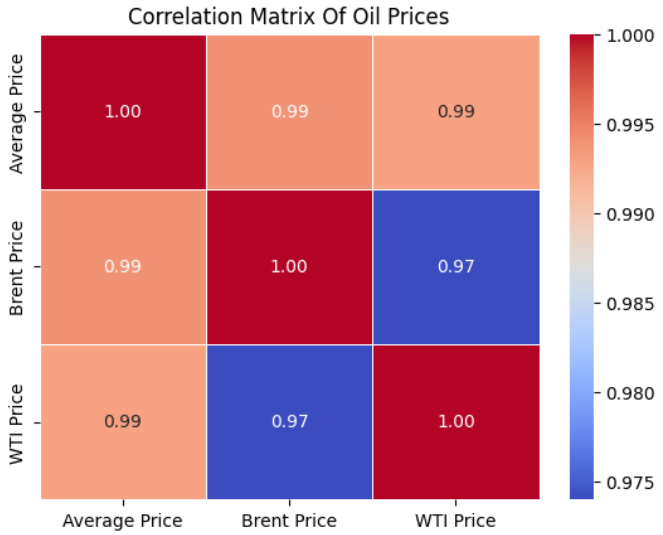


Fig. 17. Oil Prices Forecasting Features Correlation Matrice

This matrix shows that there is, a strong positive correlation between the prices of Brent and WIT oil. This is expected since the study combined the two positively correlated specifications to produce more accurate data. There is an inherent positive correlation between both specifications and the average oil price.

C. Bakery Sales Forecasting Features

1) Day: The day of the week of the date in the same row.

2) Place: the city where the transaction data is collected in the same row.

3) Date: the date on which the transaction took place in the same row.

4) isHoliday: includes the days that are holidays for the mentioned places. (The dates of the holidays are mentioned in the methods section.) Weekend holidays are not included in this holiday. If that day is a holiday, isHoliday=1, otherwise isHoliday=0

5) Brent Price: see B. Oil Price Forecasting Features part.

6) WTI Price: see B. Oil Price Forecasting Features part.

7) Avg_Oil_Price: the arithmetic average of the Brent oil price and West Texas Intermediate oil price for the same date which is present on the row.

8) Latitude: latitude of the place in that row

9) Longitude: longitude of the place in that row

10) tmin: see A. Weather Forecasting Features part.

11) tmax: see A. Weather Forecasting Features part.

12) tavg: see A. Weather Forecasting Features part.

13) wspd: see A. Weather Forecasting Features part.

14) pres: see A. Weather Forecasting Features part.

a) Correlation Matrice:

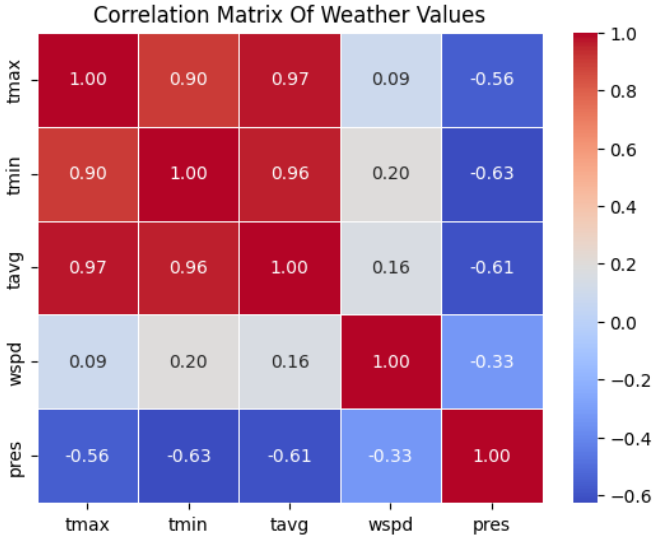


Fig. 18. Weather Forecasting Features Correlation Matrice For Bakery Sales

As can be seen from this matrix, tmin, tmax, tavg are actually highly positively correlated to each other by nature. All 3 features indicate air temperature. This is an expected situation, three features with high correlation to each other were used together in the study in order to create more sensitive data. Pressure is negatively correlated to temperature. While it can be said that wind speed is negatively correlated to pressure, wind speed is positively correlated to temperature, but this correlation is not a tight correlation.

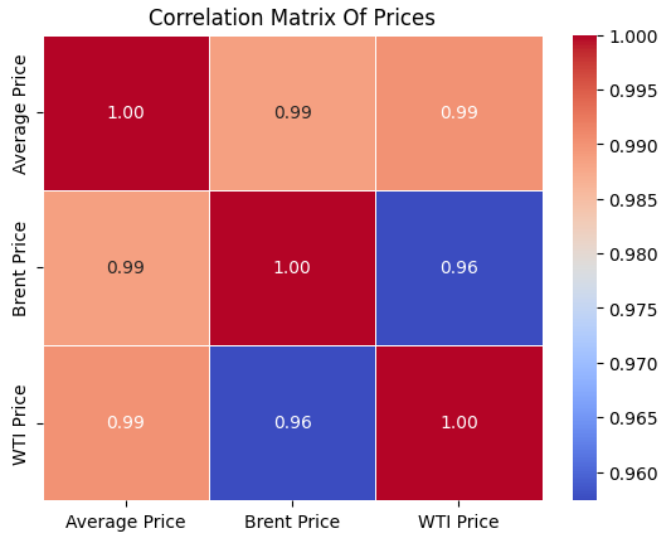


Fig. 19. Oil Prices Forecasting Features Correlation Matrice For Bakery Sales

As can be seen from this matrix, Brent and WIT oil prices are actually quite positively correlated with each other. This is to be expected, as the two positively correlated specifications were used together in the study to create more precise data. The average oil price is inherently quite positively correlated with both specifications.

The correlation matrix presents correlation relationships related to weather, oil prices and bakery sales. The relationships between temperature and oil price have been examined in detail in previous matrices. In this matrice, the focus is on the correlation of bakery sales. The correlation of each product with another is seen in the matrix. The highest correlation was observed between bread-filter coffee, tea-bread, cookie-filter coffee, while the lowest correlation was observed between filter coffee-angbutter, bread - angbutter pairs. The most neutral correlation was observed between croissant-americano pairs. (see Figure 20)

V. Experimental Results

A. Weather Forecasting Experimental Results

LSTM model was trained with Weather Forecasting Dataset prepared in this study using batch size was 32, epoch number was 50, activation function was ReLU, and loss function was Huber loss parameters. As a result of this train, metrics obtained as Huber loss 2.2357239723205566, RMSE 4.013355255126953 and R^2 0.6774458289146423.

B. Oil Price Forecasting Experimental Results

GRU model was trained with Oil Price Forecasting Dataset prepared in this study using batch size was 32, epoch number was 50, activation function was ReLU, and loss function was Mean Squared Error (MSE) parameters. As a result of this train, metrics obtained as MSE loss 189.8479766845703 RMSE 13.511139869689941, MAPE 15.606494903564453 and R^2 0.6569123268127441.

C. Bakery Sales Forecasting Experimental Results

Multiple Linear Regression model was trained with Bakery Sales Dataset prepared in this study, metrics which is optained for each product are included in APPENDIX.

D. Bakery Sales for Months Experimental Results

The test results of the monthly sales probability estimates for each product are included in APPENDIX. The test results of the monthly sales probability estimates for each product are given in APPENDIX.

VI. Discussion

A. Weather Forecasting Results Discussion

When the study was conducted as mentioned, the R^2 value was found to be 0.6774 and this value shows the closeness of the model's predictions to the true values. RMSE (4.013) shows the average distance between the model's predictions and the true values. 4 is an acceptable value here. Huber loss (2.2357) shows the amount of error in the model here. 2 is an acceptable amount of error in weather forecasts. When all these are evaluated, it is seen that the first results obtained from the model are promising, although not perfect. In addition, it is suggested that the performance of the model can be improved with various methods suggested in the future work section.

B. Oil Price Forecasting Results Discussion

When the study was conducted as mentioned, the R^2 value was found to be 0.6569 and this value shows the closeness of the model's predictions to the true values. RMSE (13.511) shows the average distance between the model's predictions and the true values. MAPE (15.606), While MAPE (15.606) shows that the model makes a prediction with a 15.606% error rate, MSE loss 189.848 calculates the error amount of the model in terms of the squared distance of the predicted values from the true values. When all these are evaluated, it is seen that the first results obtained from the model are relatively high and not perfect. In addition, it is suggested that the performance of the model can be improved with various methods suggested in the future work section.

C. Bakery Sales Forecasting Results Discussion

Best predicted products:

- Adjusted R^2: filter coffee (0.2873)
- RMSE: baguette bread (0.2712)
- RMSE Percentage: vegan mincepie (92.41%)
- MAPE: americano (80.02%).
- p-value: milk tea (0.3502)
- F-value: americano (0.8864)

Worst predicted products:

- Adjusted R^2 americano (-0.0008)
- RMSE for filter coffee (8.4624)
- RMSE for filter coffee (246.75%)
- MAPE for filter coffee (59.13%).
- p-value: americano (0.5367)
- F-value: filter coffee (56.5861)

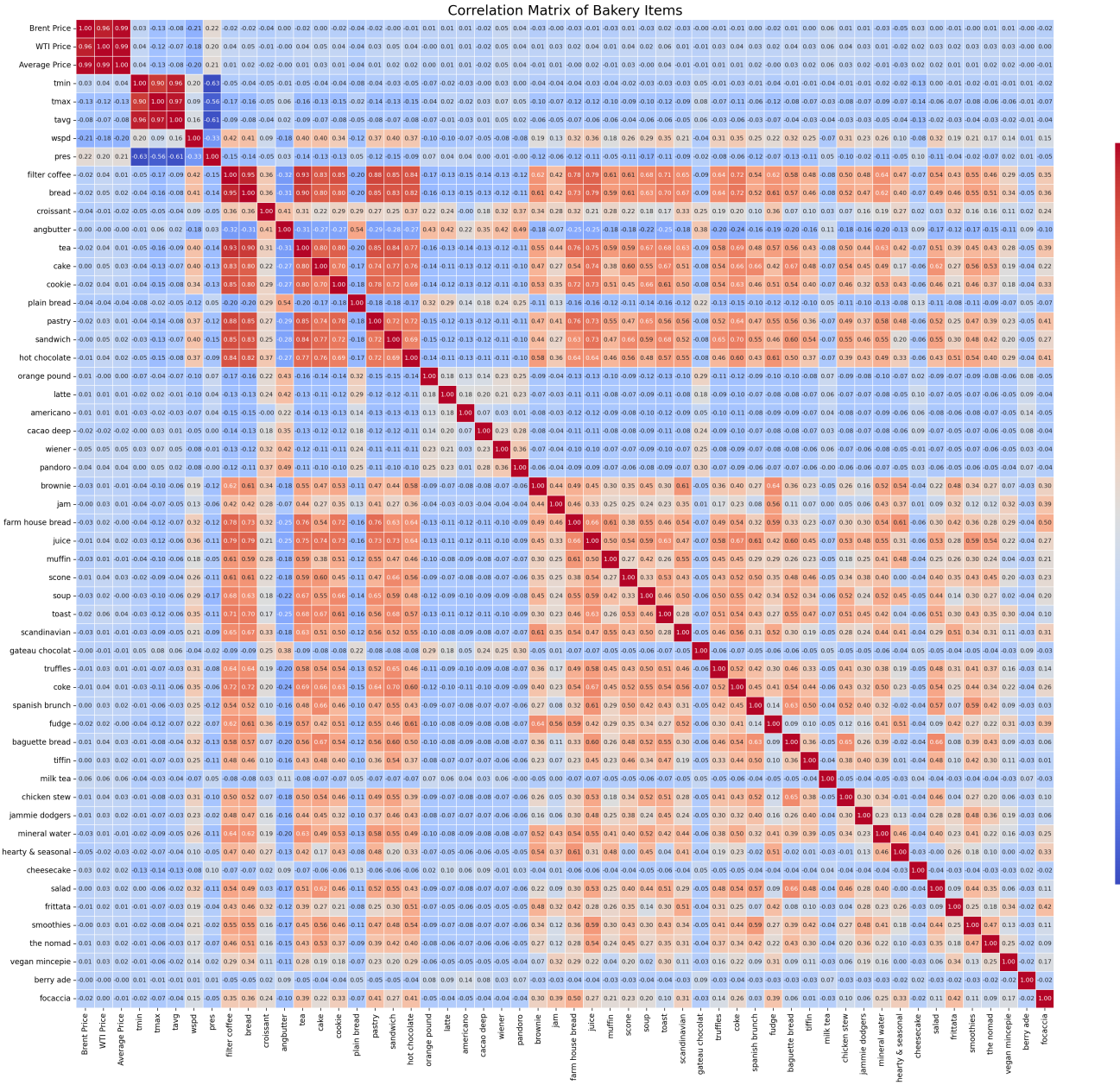


Fig. 20. Bakery Sales Features Correlation Matris of The All Features

D. Bakery Sales for Months Results Discussion

The best estimated product among these estimates is: coke (Adjusted R^2: 0.9844, F-value: 78.0750, P-value: 0.0127, RMSE: 6.5070, RMSE Percentage: 54.22%, MAPE: 20.90%). Worst predicted products: cookie (Adjusted R^2: 0.5768, F-value: 2.6655, P-value: 0.3026, RMSE: 163.9861, RMSE Percentage: 166.48%, MAPE: 535.68%), soup (Adjusted R^2: -0.3416, F-value: 0.6888, P-value: 0.7159, RMSE: 82.0520, RMSE Percentage: 193.06%, MAPE: 136.18%) and pandoro (Adjusted R^2: 0.0262, F-value: 1.0329, P-value: 0.5839, RMSE: 9.2893, RMSE Percentage: 51.61%, MAPE: 28.28%).

VII. Conclusion

In conclusion, this study proposes a prediction system to help bakeries forecast their daily and monthly sales. Bakeries struggle to forecast their sales due to the high demand for fresh products. To generate sales forecasts, the proposed method combines weather and economic variables using a variety of machine learning and deep learning techniques. Furthermore, the scope of this study includes weather and oil price forecasts for using them in bakery sales prediction. The Metostat library provided the dataset used to train the LSTM model for weather forecasting, the yfinance library provided the dataset used to train the GRU model for oil price prediction, and

Kaggle's preprocessed datasets were used to train the Multiple Linear Regression model for bakery sales prediction. This section of the study discusses the dataset preparation process, some dataset-related analyses, model outputs, and the RMSE, MAPE, R2, and Adjusted R2 scores used to evaluate the models' performance.

As can be seen from the results given in the Discussion section, the models have promising and relatively less promising results. The suggestions for improving these results are given in the Future Work section, and the studies to be carried out in continuation of this study are also included.

VIII. Future Work

A. Future Work: Next Steps for This Study

In the continuation of this study, hyperparameter finetuning will be carried out on the models used for the train up to this stage.

1) Weather Forecasting Future Work: In the LSTM model used for weather forecasting, various learning rates, batch sizes, epoch numbers, activation functions, and loss functions will be tried to improve the performance of the model, and the best performing model will be suggested among these improvements. Planned finetunings:

- Learning Rate: 0.001, 0.01, 0.1, 1
- Batch Size: 8, 16, 32, 64, 128
- Epoch Number: 10, 25, 50, 100, 200, 400, 500
- Activation Functions: ReLU, tanh, sigmoid etc.
- Loss Function: Mean Squared Error (MSE), Mean Absolute Error (MAE), Huber Loss etc.

2) Oil Price Forecasting Future Work: In the GRU model used for oil forecasting, various learning rates, batch sizes, epoch numbers, activation functions, and loss functions will be tried to improve the performance of the model, and the best performing model will be suggested among these improvements. Planned finetunings:

- Learning Rate: 0.001, 0.01, 0.1, 1
- Batch Size: 8, 16, 32, 64, 128
- Epoch Number: 10, 25, 50, 100, 200, 400, 500
- Activation Functions: ReLU, tanh, sigmoid etc.
- Loss Function: Mean Squared Error (MSE), Mean Absolute Error (MAE), Huber Loss etc.

3) Bakery Sales Forecasting Future Work: In the continuation of the study, it is planned to fine-tune by changing the solver parameter in multiple linear regression. It is also planned to try different data normalization techniques.

- Solver parameter: 'svd', 'cholesky', 'lsqr' etc.
- Data normalization: MinMaxScaler, RobustScaler etc.

B. Future Work: Recommendations for Further Research

1) Weather Forecasting Future Work:

- In this study, prediction was made using only Edinburgh data between 2015.09.07 - 2024.11.08. In further studies, a more comprehensive dataset can be created using data from wider dates and different locations.

- In this study, only the Meteostat library was used as a source and accordingly, the features obtained from this library were used. In further studies, alternative features can be obtained from different libraries. In addition, data quality can be increased by using values obtained from different sources for the same date together.

- In this study, only the LSTM model was used for weather forecasting. In other studies, especially in this study, the GRU used for oil price prediction or the widely used ARIMA model for another time series data can be used.

2) Oil Price Forecasting Future Work:

- In this study, only Brent. oil and WTI oil data between the dates 2000.08.23 - 2024.07.29 for WTI oil, 2007.08.30 - 2024.07.29 for Brent oil, 2007.08.30 - 2024.07.29 for average oil price. were used to make predictions. In further studies, a more comprehensive dataset can be created using data from wider dates and different economic indicators (gold, silver, cryptocurrency, company stock market data, dollar/euro exchange rate, etc.).

- Only the yfinance library was used as a source in this study. In further studies, data can be obtained from different libraries, datasets or stock market tracking sites. In addition, data quality can be increased by using values obtained from different sources for the same date together.

- Only the GRU model was used for oil price forecasting in this study. In other studies, the LSTM used for weather price prediction in this study or the ARIMA model widely used for another time series data can be used.

3) Bakery Sales Forecasting Future Work:

- In this study, prediction was made using only the Bakery Sales Dataset (Edinburgh) dataset and Bakery Sales (Korea) dataset obtained from Kaggle. In further studies, a more comprehensive dataset can be created using larger datasets.

- In future studies, instead of using ready-made datasets, comprehensive datasets specific to the study can be created from the sales databases of large chain bakeries.

- In this study, due to dataset limitations, only the data from a few cities in Korea and Edinburgh locations were used. This study can be expanded for different locations in future studies.

- In this study, features that are more difficult to establish and analyze such as the culture and habits of countries were not used. In a more comprehensive study, local features can be included in the study.

- In this study, only the Multiple Linear Regression model was used for Bakery Sales forecasting. In other studies, much more advanced models such as LSTM used for weather price prediction in this study, GRU used for oil price prediction, Decision Tree or Neural Networks can be used.

- A study similar to this study can also be created to predict the sales of sectors that need fresh product sales, such as grocery stores or restaurants.

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APPENDIX

Item	Adjusted R^2	F-value	P-value	RMSE	RMSE Percentage	MAPE
filter coffee	0.2873	56.5861	0.0000	8.4624	246.75%	59.13%
bread	0.2732	52.8304	0.0000	4.6929	254.20%	54.19%
croissant	0.0267	4.7820	0.0000	1.8493	103.77%	53.38%
angbutter	0.0439	7.3244	0.0000	1.0333	91.59%	34.30%
tea	0.2612	49.7480	0.0000	1.7940	247.66%	48.12%
cake	0.2263	41.3322	0.0000	1.8862	338.22%	55.95%
cookie	0.2060	36.7681	0.0000	1.4123	289.90%	68.74%
plain bread	0.0261	4.6975	0.0000	0.8023	145.54%	52.01%
pastry	0.2104	37.7493	0.0000	1.1352	310.68%	52.87%
sandwich	0.2322	42.6907	0.0000	1.2640	273.86%	51.89%
hot chocolate	0.2166	39.1164	0.0000	1.3965	315.74%	68.92%
orange pound	0.0193	3.7151	0.0001	0.5601	194.17%	67.79%
latte	0.0096	2.3406	0.0129	0.4699	209.46%	76.27%
americano	-0.0008	0.8864	0.5367	0.4465	211.06%	80.02%
cacao deep	-0.0001	0.9843	0.4511	0.5144	222.89%	80.75%
wiener	0.0219	4.0919	0.0000	0.4776	212.85%	78.64%
pandoro	0.0190	3.6695	0.0002	0.5619	194.80%	79.70%
brownie	0.0908	14.7647	0.0000	1.0176	481.04%	53.67%
jam	0.0367	6.2537	0.0000	0.4326	232.73%	73.90%
farm house bread	0.1678	28.7948	0.0000	0.5367	322.02%	53.44%
juice	0.1913	33.6105	0.0000	0.6455	359.63%	66.05%
muffin	0.0557	9.1321	0.0000	1.0243	371.59%	69.74%
scone	0.1245	20.6004	0.0000	0.8609	610.47%	63.21%
soup	0.1610	27.4566	0.0000	0.9541	647.11%	73.37%
toast	0.1809	31.4458	0.0000	0.7538	317.81%	66.61%
scandinavian	0.0781	12.6863	0.0000	0.6267	488.81%	69.51%
gateau chocolat	0.0068	1.9457	0.0423	0.3832	314.63%	89.73%
truffles	0.1235	20.4274	0.0000	0.5292	412.75%	72.73%
coke	0.1590	27.0764	0.0000	0.4165	433.13%	67.38%
spanish brunch	0.1010	16.4839	0.0000	0.3964	772.92%	65.13%
fudge	0.1008	16.4646	0.0000	0.4630	481.55%	79.29%
baguette bread	0.1292	21.4537	0.0000	0.2712	604.29%	71.76%
tiffin	0.0951	15.4947	0.0000	0.8088	573.53%	89.50%
milk tea	0.0008	1.1128	0.3502	0.3310	303.76%	91.24%
chicken stew	0.1251	20.7093	0.0000	0.3134	698.49%	81.02%
jammie dodgers	0.0754	12.2421	0.0000	0.6096	528.33%	83.93%
mineral water	0.0972	15.8518	0.0000	0.2842	443.38%	69.76%
hearty & seasonal	0.0389	6.5851	0.0000	0.6884	565.22%	90.28%
cheesecake	0.0170	3.3862	0.0004	0.2764	392.00%	91.91%
salad	0.1196	19.7401	0.0000	0.2821	628.73%	85.83%
frittata	0.0567	9.2871	0.0000	0.3939	768.02%	90.73%
smoothies	0.0638	10.4040	0.0000	0.2614	509.81%	84.16%
the nomad	0.0426	7.1341	0.0000	0.3719	644.56%	89.60%
vegan mincepie	0.0330	5.7071	0.0000	0.2184	851.79%	91.37%
berry ade	0.0094	2.3123	0.0141	0.2051	456.98%	83.25%
hocolate	0.0000	0.4892	0.6932	0.1156	432.01%	92.41%
strawberry short	0.0605	10.0590	0.0000	0.2450	442.07%	85.55%

TABLE I: Evaluation metric results of daily sales forecasts

Item	Adjusted R^2	F-value	P-value	RMSE	RMSE Percentage	MAPE
filter coffee	0.8537	8.1292	0.1143	562.2891	114.99%	81.16%
bread	0.8921	11.1086	0.0853	293.0702	102.29%	71.64%
croissant	0.7881	5.5470	0.1620	169.5408	70.06%	75.15%
angbutter	0.8416	7.4939	0.1232	149.0616	110.83%	76.69%
tea	0.7349	4.3881	0.1993	143.8110	110.20%	76.05%
cake	0.5459	2.4694	0.3214	75.3518	162.05%	114.24%
cookie	0.5768	2.6655	0.3026	163.9861	166.48%	535.68%
plain bread	0.7522	4.7099	0.1874	82.5768	104.53%	73.45%
pastry	0.8777	9.7694	0.0963	114.4834	128.63%	90.91%
sandwich	0.8204	6.5812	0.1388	36.6011	88.20%	60.88%
hot chocolate	0.9482	23.3905	0.0417	57.8352	114.53%	80.88%
orange pound	0.8842	10.3334	0.0913	33.5316	94.46%	66.41%
latte	0.4811	2.1330	0.3598	18.8259	55.37%	38.92%
americano	0.8359	7.2236	0.1275	40.2073	141.08%	98.93%
cacao deep	0.6700	3.4813	0.2430	53.7802	228.85%	146.99%
wiener	0.7264	4.2450	0.2052	16.5918	110.61%	72.36%
pandoro	0.0262	1.0329	0.5839	9.2893	51.61%	28.28%
brownie	0.6223	3.0140	0.2739	121.3105	174.55%	121.80%
jam	0.2400	1.3859	0.4879	25.1427	70.82%	65.94%
farm house bread	0.9731	45.1428	0.0219	44.6056	96.97%	67.08%
juice	0.8041	6.0153	0.1506	16.2558	79.30%	50.49%
muffin	0.7576	4.8200	0.1836	55.4723	95.64%	53.81%
scone	0.7854	5.4736	0.1640	11.4921	inf%	nan%
soup	-0.3416	0.6888	0.7159	82.0520	193.06%	136.18%
toast	0.7875	5.5306	0.1625	21.1057	inf%	nan%
scandinavian	0.7826	5.3991	0.1660	54.4485	113.43%	78.50%
gateau chocolat	0.3518	1.6632	0.4313	30.7757	267.61%	166.64%
truffles	0.8976	11.7104	0.0811	9.7787	81.49%	5.47%
coke	0.9844	78.0750	0.0127	6.5070	54.22%	20.90%
spanish brunch	0.0120	1.0149	0.5898	15.4721	inf%	nan%
fudge	0.6178	2.9754	0.2768	30.5621	119.85%	80.34%
baguette bread	-0.0786	0.9110	0.6255	24.1009	inf%	nan%
tiffin	-0.0044	0.9947	0.5964	15.5139	517.13%	160.91%
milk tea	0.6439	3.2099	0.2601	24.4623	287.79%	195.13%
chicken stew	-0.2911	0.7244	0.7000	21.1468	inf%	nan%
jammie dodgers	0.2860	1.4896	0.4651	11.8788	339.39%	73.69%
mineral water	0.9647	34.4039	0.0286	21.7240	155.17%	105.41%
hearty & seasonal	0.9579	28.7932	0.0340	42.2746	124.34%	87.54%
cheesecake	0.8213	6.6159	0.1381	7.5852	116.70%	82.20%
salad	-0.1503	0.8403	0.6521	4.5360	inf%	nan%
frittata	0.9606	30.8310	0.0318	19.6699	231.41%	71.46%
smoothies	0.8016	5.9392	0.1524	4.4116	88.23%	17.53%
the nomad	0.6831	3.6343	0.2344	9.1439	609.59%	283.48%
vegan mincepie	0.5081	2.2623	0.3669	6.8979	165.47%	143.23%
croque monsieur	0.6799	3.6365	0.2339	13.1257	99.17%	53.85%
cheese toast	0.7949	5.3105	0.1669	17.7713	163.92%	76.87%
hotdog	0.6521	3.2675	0.2454	23.1512	95.60%	66.47%

TABLE II: Evaluation metric results of monthly sales forecasts